Undecidable word problem and linear CLF

Andrew Wood   Corentin Bodart

UIMP, June 2026

Motivation

Theorem

Let G be a finitely presented group. If G has recursive Dehn function, then G has solvable word problem.

Conjugacy Length Function

Definition

Let $G = \langle S \mid R \rangle$ be a finitely generated group. Given $u, v \in F_S$ such that $u \sim v$ in G, let $\mathrm{CL}(u, v)$ be the minimal length of a word w such that $w^{-1}uw = v$. Define the *conjugacy length function* $\mathrm{CL}_G \colon \mathbb{N} \to \mathbb{N}$ of G by

$$\mathrm{CL}_G(n) = \max\{\mathrm{CL}(u, v) \mid u \sim v, |u|_S + |v|_S \leq n\}.$$

Relationship with Conjugacy Problem

Theorem

Let G be a finitely generated group with solvable word problem. If G has a recursive conjugacy length function, then G has solvable conjugacy problem.

Decide conjugacy using WP and CLF

Require: Words u, v over a finite generating set of G
Ensure: Returns TRUE if u and v are conjugate in G, and FALSE otherwise

$n \leftarrow |u| + |v|$
$N \leftarrow \mathrm{CL}_G(n)$ ▷ CLF Recursive
for all words w of length at most N **do**
    if $uw = wv$ in G **then** ▷ Use the WP
        return TRUE
    end if
end for
return FALSE

Question

Question

Can we remove the hypothesis that G has solvable word problem?

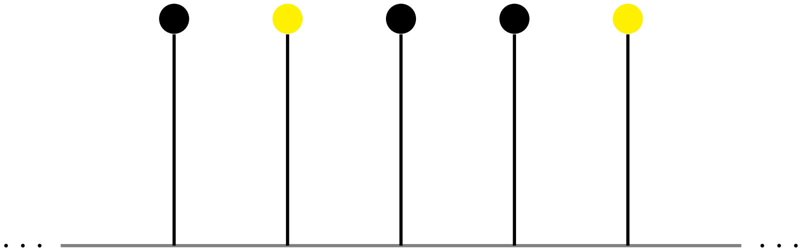
Central Extension of Lamplighter Group

Let $I \subseteq \mathbb{Z}$ with $0 \notin I$ and $I = I^{-1}$. Consider the following central extension:

$$1 \to \mathbb{Z}_2 \to G_I \to \mathbb{Z}_2 \wr \mathbb{Z} \to 1$$

It turns out G_I will have solvable word problem if, and only if, I is recursive.

Lamplighter Group



Future Work

Question

Does there exist a finitely presented group Γ with unsolvable word problem and recursive conjugacy length function?

Thank you for your attention!